Functional Graph Discriminative Network for Autism Spectrum Disorder Classification

Rohan Sinha (M25DS008)
Indian Institute of Technology Bhilai
Department of Computer Science and Engineering

Abstract

Autism spectrum disorder (ASD) is a neurodevelopmental condition associated with differences in social communication, behaviour, and brain functional organization. In this project, I reproduced the Functional Graph Discriminative Network (FGDN) approach for ASD classification using preprocessed ABIDE resting-state fMRI region-of-interest (ROI) time-series. I used AAL and HarvardOxford atlas-based ROI time-series, estimated tangent functional connectivity, constructed ASD and healthy control (HC) graph templates, and trained graph neural network models for binary ASD/HC classification. I implemented two variants: an unweighted FGDN and a weighted-template FGDN. The completed experiments used 5-fold and 10-fold cross-validation for both atlases. The best mean AUC was obtained using the AAL atlas with 10-fold weighted FGDN, achieving an AUC of 0.760592. The best mean accuracy was obtained using the HarvardOxford atlas with 10-fold unweighted FGDN, achieving an accuracy of 0.692352. An ablation study was also performed on the AAL tangent setting. The final 10-fold ablation showed that increasing the hidden dimensionality from 64 to 128 improved AUC from 0.747608 to 0.756808 compared with the reproduced AAL 10-fold unweighted baseline. The results show that tangent functional connectivity and class-specific graph templates provide a meaningful graph representation for ASD classification.

Keywords

Autism Spectrum Disorder, ABIDE, resting-state fMRI, functional connectivity, tangent embedding, graph neural network, Chebyshev graph convolution, FGDN

1 Introduction

Autism spectrum disorder (ASD) is a neurodevelopmental disorder associated with differences in social communication, restricted interests, and repetitive behaviour. Resting-state functional magnetic resonance imaging (rs-fMRI) provides a non-invasive way to study functional organization of the brain. A common approach is to divide the brain into regions of interest (ROIs), extract a time-series from each ROI, and estimate functional connectivity between ROI pairs.

In this work, I implemented a reproduction and extension of the Functional Graph Discriminative Network (FGDN) proposed by Li et al. [6]. The main idea of FGDN is to treat functional connectivity as graph-structured data. Each ROI becomes a graph node, each row of the functional connectivity matrix becomes the node feature vector, and class-specific ASD/HC templates define graph edges. A subject is passed through both ASD and HC graph branches, and the network learns which graph representation better matches the subject.

The objective of my project was to build an end-to-end ASD classification pipeline using ABIDE preprocessed ROI time-series. The final implementation includes data download, data verification, tangent functional connectivity processing, cross-validation split generation, graph template construction, PyTorch Geometric dataset construction, FGDN model training, weighted-template FGDN training, evaluation, visualization, and ablation analysis.

The source code for this project is available at: https://github.com/RohanSinha000821/FGDN-Project.

2 Dataset

2.1 ABIDE rs-fMRI ROI Time-Series

I used the Autism Brain Imaging Data Exchange (ABIDE) preprocessed resting-state fMRI ROI time-series [3]. The original FGDN paper used ABIDE data for ASD vs HC classification and evaluated the model with 5-fold and 10-fold cross-validation [6]. In this project, I used atlas-level ROI time-series rather than raw 4D fMRI volumes.

Each subject is represented by a numerical ROI time-series matrix:

$$X_i \in \mathbb{R}^{T_i \times N}, \tag{1}$$

where T_i is the number of time points for subject i, and N is the number of ROIs in the atlas. The rows are time points and the columns are ROI signals.

The two completed atlases are summarized in Table 1. Figure 1 visualizes the subject and ROI counts.

Table 1: Dataset summary used in the completed project.

Atlas	Matched Subjects	ROIs
AAL	531	116
HarvardOxford	884	111

2.2 Labels

The ABIDE phenotypic file provides the diagnosis group through `DX_GROUP`. In my implementation, I used the binary label convention:

$$\text{HC} = 0, \qquad \text{ASD} = 1. \tag{2}$$

The classification task is therefore a binary prediction problem.

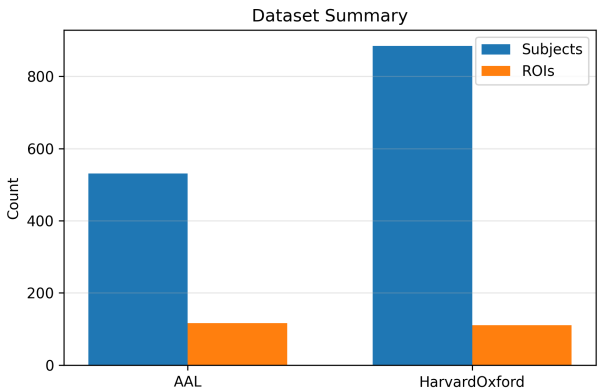

Figure 1: Dataset summary for AAL and HarvardOxford.

3 Functional Connectivity

3.1 From ROI Time-Series to FC Matrix

For each subject, the ROI time-series matrix is converted into a functional connectivity matrix:

$$F_i \in \mathbb{R}^{N \times N}. \tag{3}$$

Each element $F_i(p, q)$ represents the functional relationship between ROI p and ROI q.

In a simple correlation-based approach, the entry $F_i(p, q)$ can be interpreted as the correlation between two ROI time-series. In this project, I used tangent-space functional connectivity, because the original FGDN paper reports strong performance with tangent features [6], and tangent embedding has also been used in functional connectome benchmarking [1].

3.2 Tangent Functional Connectivity

Tangent functional connectivity represents each subject's covariance structure in a tangent space around a group reference covariance. Let Σ_i denote the covariance matrix estimated for subject i. Tangent embedding maps subject covariance matrices to a common tangent space:

$$T_i = \log_{\bar{\Sigma}}(\Sigma_i), \tag{4}$$

where $\bar{\Sigma}$ is a group reference covariance and $\log_{\bar{\Sigma}}(\cdot)$ denotes the logarithmic map at the reference point.

In implementation, I used Nilearn's ConnectivityMeasure (kind="tangent") with the Ledoit-Wolf shrinkage covariance estimator [5]. For each cross-validation fold, the tangent estimator was fitted on the training subjects of that fold and then used to transform both training and test subjects. This produced fold-specific tangent connectivity matrices.

4 Graph Representation

4.1 Functional Connectivity as a Graph

A graph is defined as:

$$G = (V, E, W), \tag{5}$$

where V is the set of nodes, E is the set of edges, and W is the adjacency matrix.

In this project:

- each ROI is one graph node;
- each row of the subject connectivity matrix is the feature vector of one node;
- ASD and HC templates define class-specific graph edges.

For a subject connectivity matrix $F_i \in \mathbb{R}^{N \times N}$, the node feature matrix is:

$$X_i = F_i. \tag{6}$$

Thus, the feature of node p is the p-th row of the functional connectivity matrix:

$$x_p = F_i[p, :]. \tag{7}$$

4.2 ASD and HC Graph Templates

For each fold, I computed the mean connectivity matrix separately for ASD and HC training samples:

$$\bar{F}_{ASD} = \frac{1}{N_{ASD}} \sum_{i:y_i=1} F_i, \tag{8}$$

$$\bar{F}_{HC} = \frac{1}{N_{HC}} \sum_{i:y_i=0} F_i. \tag{9}$$

The rows of these class mean matrices are treated as ROI feature vectors for template construction. A K-nearest-neighbour rule with $k = 20$ is applied to form class-specific adjacency matrices:

$$A_{ASD}, \qquad A_{HC}. \tag{10}$$

A subject therefore produces two graph views:

$$G_{ASD} = (X_i, A_{ASD}), \qquad G_{HC} = (X_i, A_{HC}). \tag{11}$$

Both graphs have the same node features but different class-specific template edges. This follows the graph construction idea used in the original FGDN paper, where functional connectivity is combined with ASD and HC templates to form two graphs [6].

5 Graph Convolutional Networks

5.1 Why Graph Convolution Is Needed

A standard convolutional neural network assumes that data lie on a regular grid, such as pixels in an image. Brain connectivity graphs are not regular grids. ROIs form nodes, and edges are defined by functional similarity or template structure. Therefore, a graph convolutional network is more suitable than a standard CNN.

A graph convolution layer updates node features by aggregating information from neighbouring nodes. A general graph propagation layer can be written as:

$$H^{(l+1)} = \sigma\left(\mathcal{G}(A) H^{(l)} \Theta^{(l)}\right), \tag{12}$$

where $H^{(l)}$ is the node representation at layer l, A is the adjacency matrix, $\Theta^{(l)}$ is a learnable weight matrix, σ is a non-linear activation function, and $\mathcal{G}(A)$ is a graph-based propagation operator.

5.2 Spectral Graph Convolution

Spectral graph convolution is defined using the graph Laplacian. For a graph with adjacency matrix W and degree matrix D, the normalized graph Laplacian is:

$$L = I_N - D^{-1/2} W D^{-1/2}. \tag{13}$$

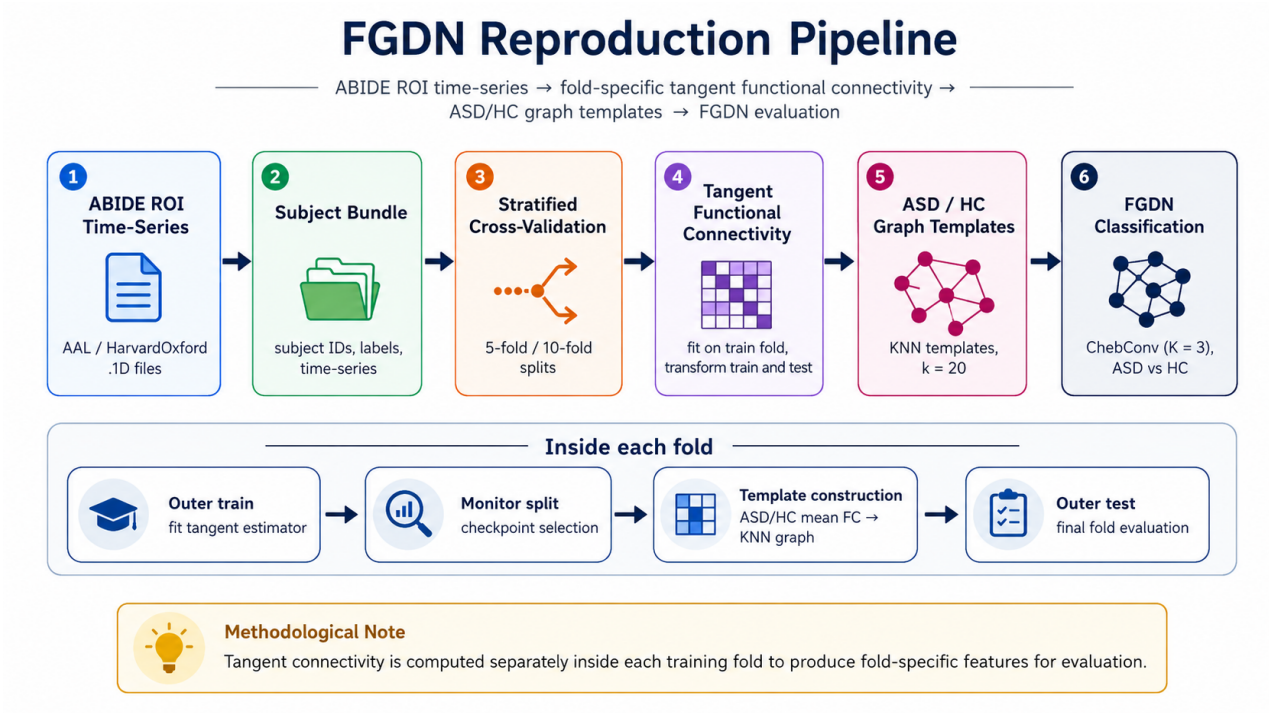

Figure 2: End-to-end FGDN reproduction pipeline used in this project. ROI time-series are converted into subject bundles, split by cross-validation, transformed into fold-specific tangent connectivity matrices, converted into ASD/HC KNN graph templates, and used for FGDN training and test evaluation.

The eigendecomposition of L is:

$$L = U\Lambda U^T, \tag{14}$$

where U contains eigenvectors and Λ contains eigenvalues. For a graph signal x, spectral graph convolution with a filter g_θ is:

$$g_\theta *_G x = U g_\theta(\Lambda) U^T x. \tag{15}$$

The limitation of direct spectral graph convolution is that eigendecomposition is computationally expensive and graph-specific. This motivates Chebyshev polynomial approximation.

5.3 Chebyshev Graph Convolution

Chebyshev graph convolution approximates the spectral filter using Chebyshev polynomials, following the localized spectral filtering approach proposed by Defferrard et al. [2]. The filter is approximated as:

$$g_\theta(\Lambda) \approx \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\Lambda}), \tag{16}$$

where T_k is the k-th Chebyshev polynomial and $\tilde{\Lambda}$ is the rescaled eigenvalue matrix. In matrix form, the convolution becomes:

$$g_\theta *_G x \approx \sum_{k=0}^{K-1} \theta_k T_k(\tilde{L}) x, \tag{17}$$

where:

$$\tilde{L} = \frac{2L}{\lambda_{max}} - I_N. \tag{18}$$

The Chebyshev polynomials are recursively defined as:

$$T_0(x) = 1, \tag{19}$$

$$T_1(x) = x, \tag{20}$$

$$T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x), \quad k \geq 2. \tag{21}$$

I used Chebyshev graph convolution because it provides localized spectral filtering and avoids direct graph Fourier-basis computation. In the baseline configuration, the Chebyshev order was set to $K = 3$, following the FGDN paper configuration [6]. The ablation study later evaluated the effect of changing this order.

6 FGDN Architecture

The FGDN model uses two class-specific graph branches. The same subject functional connectivity matrix is represented once using an ASD graph template and once using an HC graph template. These two graph views are processed by graph convolutional layers, then mapped to class scores for ASD/HC prediction. Figure 3 shows the reference architecture from the original FGDN paper.

6.1 Unweighted FGDN

The implemented unweighted FGDN has two branches:

- an ASD graph branch using the ASD template adjacency;
- an HC graph branch using the HC template adjacency.

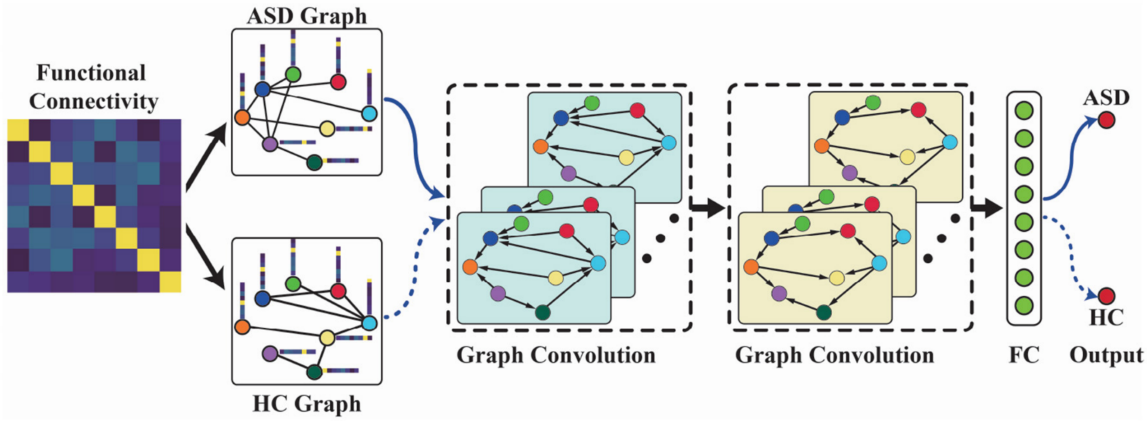

Figure 3: Functional Graph Discriminative Network architecture proposed by Li et al. [6]. The model constructs ASD and HC functional graphs from a subject's functional connectivity matrix, processes them using graph convolutional layers, and predicts ASD or HC through a fully connected output layer.

Each branch has the structure:

$$\text{ChebConv} \rightarrow \text{PReLU} \rightarrow \text{Dropout} \rightarrow \text{ChebConv} \rightarrow \text{PReLU} \rightarrow \text{Dropout}. \tag{22}$$

The final node features are flattened and passed through a linear layer to produce a scalar branch score. The two branch scores are combined to produce binary class logits.

6.2 Weighted FGDN

I also implemented a weighted-template extension. The unweighted model uses binary template adjacency, whereas the weighted model passes edge weights into the Chebyshev convolution operation.

The weighted graph uses Gaussian-style KNN weights based on distances between ROI row-vectors in the class mean functional connectivity matrix:

$$w_{ij} = \exp\left(-\frac{d(i,j)^2}{2\theta^2}\right), \tag{23}$$

where $d(i, j)$ is the distance between ROI feature vectors and θ is estimated from the mean non-zero KNN distance. This weighted version was evaluated as an extension of the original FGDN reproduction.

7 Implementation Workflow

The project was implemented as a modular Python pipeline:

(1) download ABIDE phenotypic and ROI time-series data;
(2) verify subject IDs, labels, and ROI dimensions;
(3) build subject time-series bundles;
(4) create stratified 5-fold and 10-fold splits;
(5) compute fold-specific tangent connectivity;
(6) build ASD and HC KNN graph templates;
(7) build PyTorch Geometric datasets [4];
(8) train unweighted FGDN;
(9) evaluate unweighted FGDN;
(10) train weighted FGDN;
(11) evaluate weighted FGDN;
(12) run ablation studies;
(13) summarize results.

8 Experimental Setup

The baseline experiments used the hyperparameters shown in Table 2. These values follow the original FGDN setting where possible [6].

Table 2: Baseline training and model hyperparameters.

Parameter	Value
KNN neighbours for template	20
Chebyshev order	3
Hidden channels	64
Dropout	0.1
Batch size	16
Learning rate	0.0001
Weight decay	0.0005
Optimizer	Adam
Epochs	100
Patience	20
Monitor split	10% of outer training fold

These values define the baseline configuration. The ablation study later varied template size, Chebyshev order, hidden channels, dropout, learning rate, weight decay, batch size, and weighted versus binary graph templates.

The model was evaluated using both 5-fold and 10-fold cross-validation. I measured mean accuracy and mean AUC across folds.

9 Results

9.1 Cross-Validation Results

Table 3 shows the final results obtained in the completed experiments.

Table 3: Final FGDN results obtained in this project.

Variant	Atlas / Fold Setting	Accuracy Mean	Accuracy SD	AUC Mean	AUC SD
Unweighted	AAL 5-fold	0.657150	0.052083	0.733335	0.039354
Unweighted	AAL 10-fold	0.662858	0.058932	0.747608	0.066337
Unweighted	HarvardOxford 5-fold	0.671970	0.016687	0.732362	0.011817
Unweighted	HarvardOxford 10-fold	**0.692352**	0.050286	0.739556	0.038684
Weighted	AAL 5-fold	0.664750	0.042253	0.746564	0.031682
Weighted	AAL 10-fold	0.670335	0.038649	**0.760592**	0.061851
Weighted	HarvardOxford 5-fold	0.656112	0.030566	0.722158	0.023552
Weighted	HarvardOxford 10-fold	0.677605	0.034057	0.744680	0.032152

The best overall AUC was obtained by the AAL 10-fold weighted FGDN model, with an AUC of 0.760592. The best overall accuracy was obtained by the HarvardOxford 10-fold unweighted FGDN model, with an accuracy of 0.692352.

9.2 Comparison With the Original FGDN Paper

Table 4 compares the results obtained in this project with the FGDN results reported by Li et al. [6]. The original paper reported results as percentages; therefore, the project results are also shown as percentages in this comparison table.

The original FGDN paper reported stronger AAL and HO tangent results than this project. For AAL 10-fold tangent FGDN, Li et al. [6] reported 70.8% accuracy and 79.4% AUC, whereas the reproduced unweighted AAL 10-fold model in this project achieved 66.29% accuracy and 74.76% AUC. The weighted AAL 10-fold extension improved the reproduced AUC to 76.06%, but it still remained below the original paper's reported AAL tangent AUC.

Several factors make this comparison non-identical. The original paper used 866 ABIDE samples and also reported MODL results, where MODL tangent achieved the best 5-fold result. In this project, the completed datasets were AAL with 531 matched subjects and HarvardOxford with 884 matched subjects. Therefore, the comparison should be interpreted as a reproduction-oriented comparison rather than an exact replication under identical data conditions.

9.3 Ablation Study

To study the effect of major model and graph-construction choices, I performed an ablation study on the AAL tangent setting. The ablation study was conducted in three stages.

First, I ran a quick exploratory ablation on AAL 5-fold fold 1. This stage varied the KNN template size, Chebyshev order, dropout, hidden channels, learning rate, weight decay, batch size, and weighted versus unweighted template construction. Second, the most promising settings from the exploratory run were evaluated using full AAL 5-fold cross-validation. Third, the two strongest candidates were evaluated using full AAL 10-fold cross-validation and compared against the original AAL 10-fold unweighted baseline.

9.3.1 Exploratory Fold-1 Ablation. Table 5 summarizes the best setting from each exploratory ablation group on AAL 5-fold fold 1.

The fold-1 exploratory ablation suggested that graph sparsity, model capacity, and Chebyshev order could influence performance. However, since these results were obtained on only one fold, they were treated only as candidate-selection results.

9.3.2 AAL 5-Fold Confirmation Ablation. The most promising settings from the exploratory stage were evaluated using full AAL 5-fold cross-validation. The results are shown in Table 6.

The AAL 5-fold confirmation showed that template_k = 10 gave the highest mean AUC, while hidden_channels = 128 gave the highest mean accuracy. The combined setting did not outperform the simpler single-factor changes, indicating that individually useful changes were not necessarily additive.

9.3.3 AAL 10-Fold Final Ablation. Based on the 5-fold confirmation, two candidates were selected for final AAL 10-fold evaluation: template_k = 10 and hidden_channels = 128. These were compared against the original AAL 10-fold unweighted baseline. The results are shown in Table 7.

The final ablation showed that hidden_channels = 128 achieved the best AUC, improving from the reproduced AAL 10-fold baseline AUC of 0.747608 to 0.756808. The template_k = 10 setting achieved the highest accuracy, but its AUC was slightly below the baseline. Since AUC is more informative than accuracy for threshold-independent ASD/HC classification, hidden_channels = 128 was selected as the final ablation improvement.

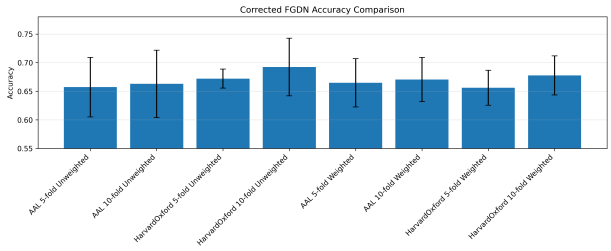

Figure 4: Accuracy comparison across atlas, fold setting, and model variant.

10 Visual Analysis

10.1 Functional Connectivity Difference

Figure 6 shows the absolute difference between ASD and HC mean functional connectivity for AAL fold 1. This visualization shows

Table 4: Comparison between original FGDN paper results and this project.

Source	Variant	Atlas	CV Setting	Accuracy (%)	AUC (%)
Li et al. [6]	FGDN	HO Tangent	5-fold	71.4	76.9
This project	Unweighted FGDN	HarvardOxford Tangent	5-fold	67.20	73.24
This project	Weighted FGDN	HarvardOxford Tangent	5-fold	65.61	72.22
Li et al. [6]	FGDN	AAL Tangent	5-fold	72.3	75.1
This project	Unweighted FGDN	AAL Tangent	5-fold	65.72	73.33
This project	Weighted FGDN	AAL Tangent	5-fold	66.48	74.66
Li et al. [6]	FGDN	HO Tangent	10-fold	71.4	77.7
This project	Unweighted FGDN	HarvardOxford Tangent	10-fold	69.24	73.96
This project	Weighted FGDN	HarvardOxford Tangent	10-fold	67.76	74.47
Li et al. [6]	FGDN	AAL Tangent	10-fold	70.8	79.4
This project	Unweighted FGDN	AAL Tangent	10-fold	66.29	74.76
This project	Weighted FGDN	AAL Tangent	10-fold	67.03	76.06
This project	Ablated FGDN	AAL Tangent, hidden=128	10-fold	67.22	75.68

Table 5: Best setting from each exploratory fold-1 ablation group on AAL tangent.

Ablation Group	Best Setting	Accuracy	AUC
KNN neighbours	template_k = 20	0.672897	**0.753497**
Dropout	dropout = 0.0	0.682243	0.752098
Hidden channels	hidden_channels = 128	**0.719626**	0.748601
Chebyshev order	cheb_k = 1	0.663551	0.747203
Weighted vs binary template	binary	0.644860	0.740909
Weight decay	0.0005	0.691589	0.739860
Learning rate	0.00005	0.691589	0.739860
Batch size	16	0.663551	0.735315

Table 6: AAL 5-fold confirmation ablation results.

Candidate	Accuracy Mean	Accuracy SD	AUC Mean	AUC SD
template_k = 10	0.655369	0.044505	**0.751116**	0.042558
hidden_channels = 128	**0.674184**	0.046615	0.750554	0.052048
combined top setting	0.672298	0.057266	0.750421	0.050429
cheb_k = 1	0.660959	0.071525	0.747211	0.054616
dropout = 0.0	0.653465	0.046079	0.745837	0.063420

Table 7: Final AAL 10-fold ablation comparison against the reproduced baseline.

Setting	Accuracy Mean	Accuracy SD	AUC Mean	AUC SD	Accuracy Gain	AUC Gain
Baseline, hidden=64, template_k=20	0.662858	0.058932	0.747608	0.066337	–	–
template_k = 10	**0.674109**	0.075850	0.746143	0.076860	+0.011251	-0.001465
hidden_channels = 128	0.672187	0.073406	**0.756808**	0.090864	+0.009329	**+0.009200**

how the class-specific mean connectivity patterns differ before graph template construction.

10.2 ROI-Level Difference Scores

Figure 7 shows the top discriminative ROI indices based on ASD/HC template difference scores for AAL. This is a qualitative ROI-level analysis based on differences in the constructed class templates. It is not the same as the MODL region-ablation analysis in the original

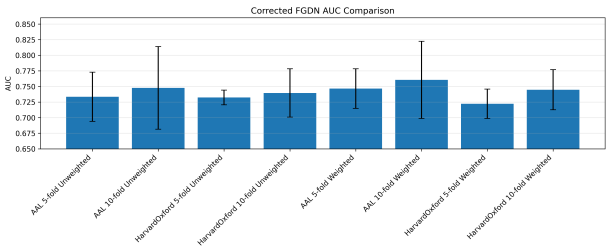

Figure 5: AUC comparison across atlas, fold setting, and model variant.

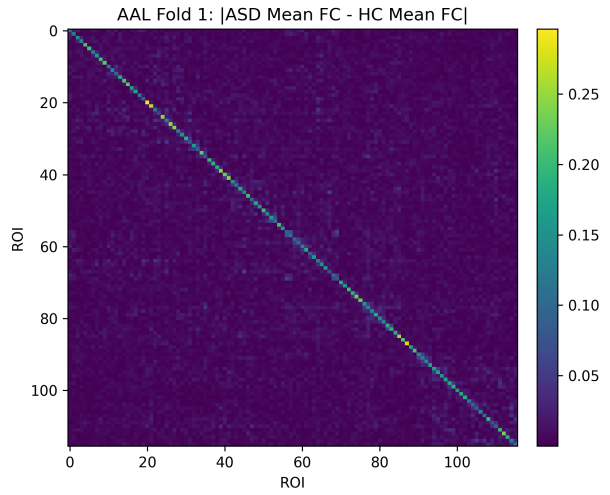

Figure 6: Absolute difference between ASD and HC mean functional connectivity for AAL fold 1.

FGDN paper, but it provides a useful visualization of which ROI indices contributed most strongly to template differences in the current AAL setup.

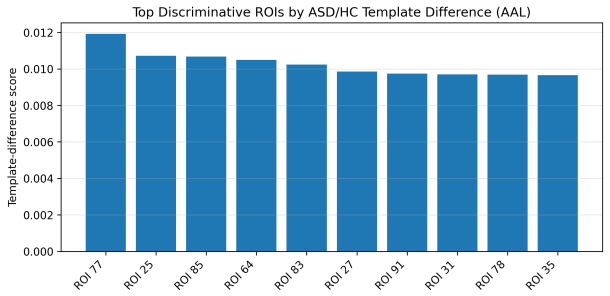

Figure 7: Top ROI indices ranked by ASD/HC template difference score for AAL.

11 Discussion

The results show that both unweighted and weighted FGDN models can learn meaningful representations from tangent functional connectivity graphs. AAL with weighted templates achieved the highest mean AUC among the main reproduction experiments, suggesting that weighted edges can preserve useful information that binary graph templates discard. HarvardOxford with unweighted templates achieved the highest mean accuracy, suggesting that the effect of weighting depends on atlas choice.

Compared with the original FGDN paper, the reproduced results in this project are lower for the directly comparable AAL and HarvardOxford tangent settings. For example, Li et al. [6] reported 70.8% accuracy and 79.4% AUC for AAL tangent 10-fold FGDN, while this project obtained 66.29% accuracy and 74.76% AUC using the reproduced unweighted AAL 10-fold model. The weighted AAL 10-fold extension improved AUC to 76.06%, narrowing the gap but still remaining below the paper's reported AAL tangent result.

The difference may be due to several implementation and data factors. The original paper used 866 ABIDE samples and evaluated AAL, HarvardOxford, and MODL atlases. In contrast, the completed project used 531 matched AAL subjects and 884 matched HarvardOxford subjects. The exact preprocessing pipeline, subject inclusion set, random cross-validation splits, and implementation details may also differ. Therefore, the project should be interpreted as a faithful reproduction and extension attempt rather than an exact numerical replication.

The ablation study provided useful insight into which design choices matter most. The exploratory fold-1 ablation suggested that template sparsity, hidden dimensionality, Chebyshev order, and regularization can affect performance. However, the full 5-fold and 10-fold confirmation stages showed that not all fold-1 improvements generalize. In the final AAL 10-fold ablation, increasing hidden channels from 64 to 128 produced the best AUC, improving the reproduced AAL 10-fold baseline from 0.747608 to 0.756808. The template_k = 10 setting gave the best accuracy, but did not improve AUC over the baseline.

The combined ablation candidate did not outperform the simpler single-factor changes. This is an important observation because it shows that individually promising hyperparameter changes do not necessarily combine additively. For this project, the simplest useful final ablation improvement was increasing the model capacity to 128 hidden channels.

The visualizations support the main methodological idea: ASD and HC groups produce different class mean functional connectivity matrices, and these matrices produce class-specific graph templates. The FGDN model learns from both graph views of the same subject.

12 Limitations

The main limitations of the completed project are:

- only AAL and HarvardOxford were completed, while the original FGDN paper also reported MODL results;
- the AAL and HarvardOxford subject counts in this project differ from the 866-subject setup reported in the original paper;
- the exact MODL discriminative brain-region analysis from the original paper was not reproduced;
- the brain-region visualization is ROI-index based rather than atlas-coordinate based;

- the current implementation focuses on static functional connectivity;
- training curves indicated possible overfitting on some folds, especially when the training AUC approached 1.0 while monitor performance saturated;
- the final ablation improvement was modest, so statistical testing would be needed before claiming a strong performance improvement.

13 Future Work

The main future extension is dynamic functional connectivity. Instead of using one static connectivity matrix per subject, rs-fMRI time-series can be divided into temporal windows, producing a sequence of connectivity matrices. A future model could combine graph convolution with temporal modelling to learn how brain connectivity changes over time. This may capture ASD-related temporal patterns that are not visible in static functional connectivity.

Other future improvements include stronger regularization, better graph sparsification, hyperparameter tuning, site-aware evaluation, and atlas-coordinate visualization for brain-region interpretability. Another useful extension would be to run the final hidden_channels = 128 ablation on HarvardOxford to test whether the model-capacity improvement generalizes across atlases.

14 Conclusion

In this project, I reproduced and extended the FGDN approach for ASD classification using ABIDE preprocessed rs-fMRI ROI time-series. I converted ROI time-series into tangent functional connectivity matrices, constructed ASD and HC graph templates, trained unweighted and weighted FGDN models, and evaluated them using 5-fold and 10-fold cross-validation.

The best overall AUC in the main experiments was obtained by AAL 10-fold weighted FGDN, with an AUC of 0.760592. The best overall accuracy was obtained by HarvardOxford 10-fold unweighted FGDN, with an accuracy of 0.692352. Compared with the original FGDN paper, the reproduced results were lower for directly comparable AAL and HarvardOxford tangent settings, likely due to differences in subject counts, preprocessing, and implementation details.

The ablation study showed that increasing the model capacity from 64 to 128 hidden channels produced the best final AUC improvement on AAL 10-fold evaluation. The reproduced AAL 10-fold baseline achieved an AUC of 0.747608, while hidden_channels = 128 achieved an AUC of 0.756808. The template_k = 10 setting achieved the highest ablation accuracy, but did not improve AUC over the baseline. Therefore, hidden_channels = 128 was selected as the final ablation improvement.

Overall, the project demonstrates how functional connectivity can be represented as graph-structured data and classified using Chebyshev graph convolutional networks. It also shows that graph construction, edge weighting, atlas choice, and model capacity all influence FGDN performance.

References

[1] Kamalaker Dadi, Mehdi Rahim, Alexandre Abraham, Darya Chyzhyk, Michael Milham, Bertrand Thirion, and Gael Varoquaux. 2019. Benchmarking functional connectome-based predictive models for resting-state fMRI. *NeuroImage* 192 (2019), 115–134. doi:10.1016/j.neuroimage.2019.02.062

[2] Michaël Defferrard, Xavier Bresson, and Pierre Vandergheynst. 2016. Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering. In *Advances in Neural Information Processing Systems*. 3844–3852.

[3] Adriana Di Martino, Chao-Gan Yan, Qingyang Li, Erin Denio, F. Xavier Castellanos, Kaat Alaerts, Jeffrey S. Anderson, Michal Assaf, Susan Y. Bookheimer, Mirella Dapretto, et al. 2014. The Autism Brain Imaging Data Exchange: towards a large-scale evaluation of the intrinsic brain architecture in autism. *Molecular Psychiatry* 19, 6 (2014), 659–667. doi:10.1038/mp.2013.78

[4] Matthias Fey and Jan Eric Lenssen. 2019. Fast Graph Representation Learning with PyTorch Geometric. In *ICLR Workshop on Representation Learning on Graphs and Manifolds*.

[5] Olivier Ledoit and Michael Wolf. 2004. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis* 88, 2 (2004), 365–411. doi:10.1016/S0047-259X(03)00096-4

[6] Jingcong Li, Fei Wang, Jiahui Pan, and Zhenfu Wen. 2021. Identification of Autism Spectrum Disorder With Functional Graph Discriminative Network. *Frontiers in Neuroscience* 15 (2021), 729937. doi:10.3389/fnins.2021.729937